Microsoft Sentinel SOC Lab

End-to-End SIEM Deployment, Data Ingestion & KQL Threat Detection

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Platform	Microsoft Azure + Microsoft Sentinel + Azure Data Explorer
Region	Central India
Workspace	sentinal-workspace (Log Analytics)
Data Connectors	8 Provisioned via Training Lab + Azure Activity manually configured
Incidents Generated	3 Live Incidents (1 High, 2 Medium Severity)
KQL Queries	4 SOC Detection Queries on Azure Data Explorer

1. Azure Free Trial Setup

Microsoft Azure offers a free trial with $200 credits valid for 30 days. This is sufficient to deploy a full Microsoft Sentinel lab at no cost.

Step 1	Go to portal.azure.com → Click 'Start Free'
Step 2	Sign in or create a Microsoft account
Step 3	Enter details — Name, Phone, Country (India)
Step 4	Add debit/credit card for identity verification (not charged)
Step 5	Complete verification — get $200 credits + 30 days access
Step 6	You land on the Azure Portal dashboard — trial is active

Important: *Do NOT cancel subscription early — cancelling immediately revokes access. Let it expire naturally.*

2. Resource Group & Log Analytics Workspace

A Resource Group is a logical container for all Azure resources. The Log Analytics Workspace is the data store for Microsoft Sentinel.

Field	Value Used
Subscription	Azure subscription 1
Resource Group	sentinal-rg (New)
Workspace Name	sentinal-workspace
Region	Central India

Navigation: Azure Portal → Search 'Log Analytics workspaces' → Create → Fill details → Review + Create

3. Microsoft Sentinel Deployment

Microsoft Sentinel is deployed on top of the Log Analytics Workspace. The Training Lab Solution auto-provisions attack simulation data and 8 data connectors.

- Search 'Microsoft Sentinel' → Create → Select sentinal-workspace → Add
- Content Hub → Search 'Microsoft Sentinel Training Lab Solution' → Install
- Result: 3 Analytic Rules, 2 Hunting Queries, 1 Playbook, 8 Data Connectors provisioned
- Azure Activity connector showed Disconnected — manually opened connector page and reconfigured it

4. Incidents Generated

Severity	Count	Status
High	1	New
Medium	2	New
Low	0	—

5. KQL Detection Queries

All queries written and tested on Azure Data Explorer (dataexplorer.azure.com) using IntroToKQL database — same KQL engine as Microsoft Sentinel.

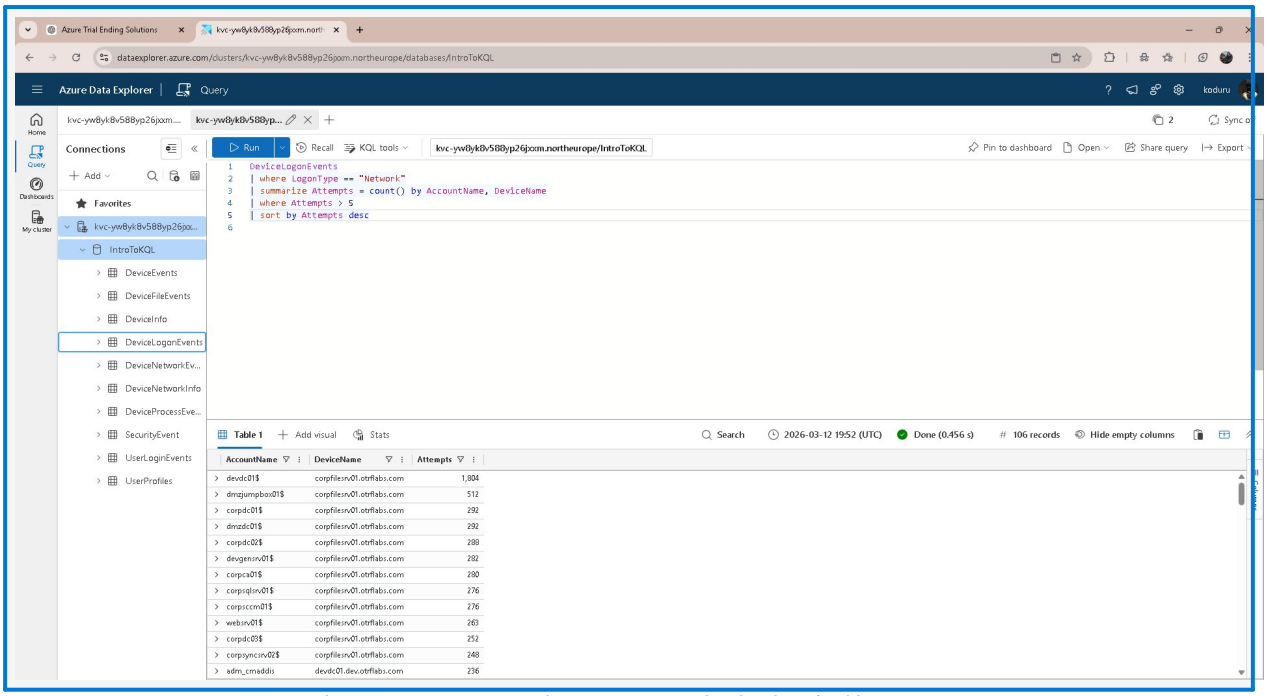
Tables used: *DeviceLogonEvents, DeviceProcessEvents, DeviceNetworkEvents, UserLoginEvents*

Query 1 — Brute Force / Network Logon Detection

Detects accounts with excessive network logon attempts. Result: 106 records, top account showed 1,804 attempts.

```
DeviceLogonEvents
| where LogonType == "Network"
| summarize Attempts = count() by AccountName, DeviceName
| where Attempts > 5
| sort by Attempts desc
```

SOC Use Case: Account compromise, credential stuffing, brute force detection



Screenshot: Brute Force Detection — 106 records, devdc01$ with 1,804 attempts

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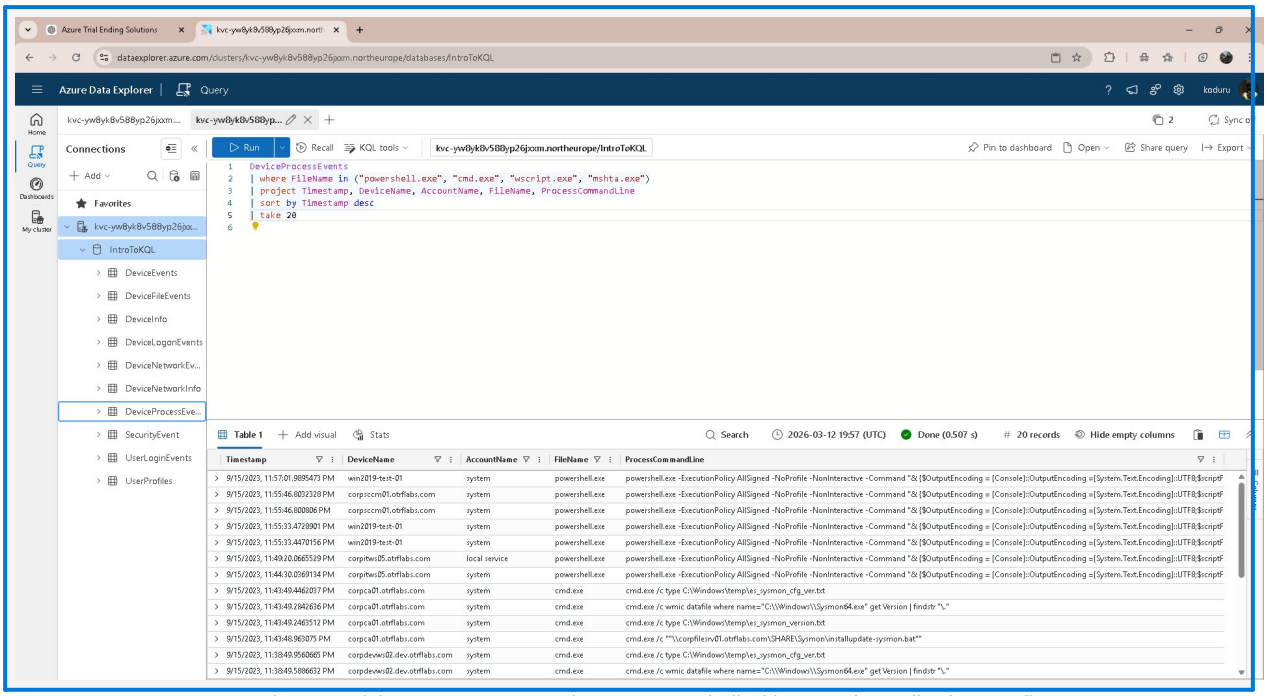
Query 2 — Suspicious Process Execution (LOLBins)

Identifies execution of PowerShell, CMD, WScript, MSHTA — common attacker tools. Result: powershell.exe with -NoProfile -NonInteractive flags visible.

```
DeviceProcessEvents
| where FileName in ("powershell.exe", "cmd.exe", "wscript.exe", "mshta.exe")
| project Timestamp, DeviceName, AccountName, FileName, ProcessCommandLine
| sort by Timestamp desc
| take 20
```

SOC Use Case: Malware execution, fileless attacks, post-exploitation activity



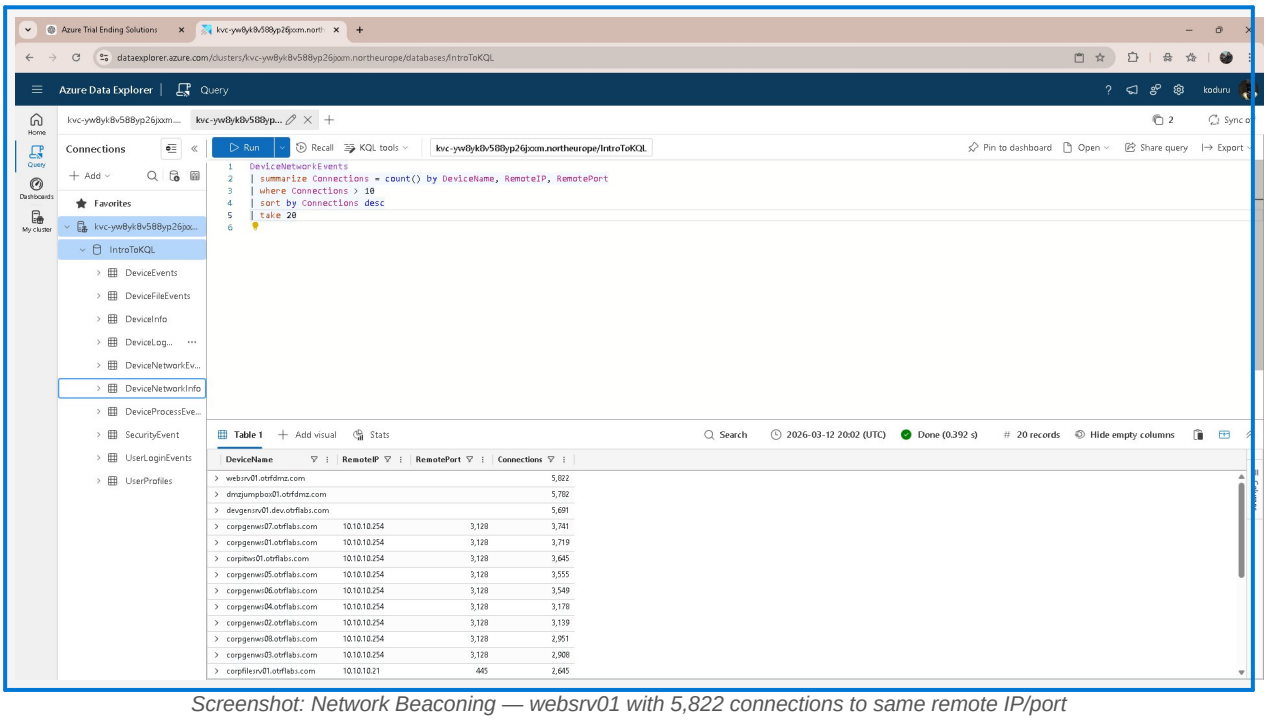
Screenshot: Suspicious Process Execution — PowerShell with execution policy bypass flags

Query 3 — Network Beaconing / C2 Detection

Detects repeated connections to same IP/port — classic C2 beaconing. Result: websrv01 made 5,822 connections; multiple devices hitting 10.10.254:3128.

```
DeviceNetworkEvents
| summarize Connections = count() by DeviceName, RemoteIP, RemotePort
| where Connections > 10
| sort by Connections desc
| take 20
```

SOC Use Case: C2 communication, data exfiltration, persistent malware traffic



Screenshot: Network Beaconing — websrv01 with 5,822 connections to same remote IP/port

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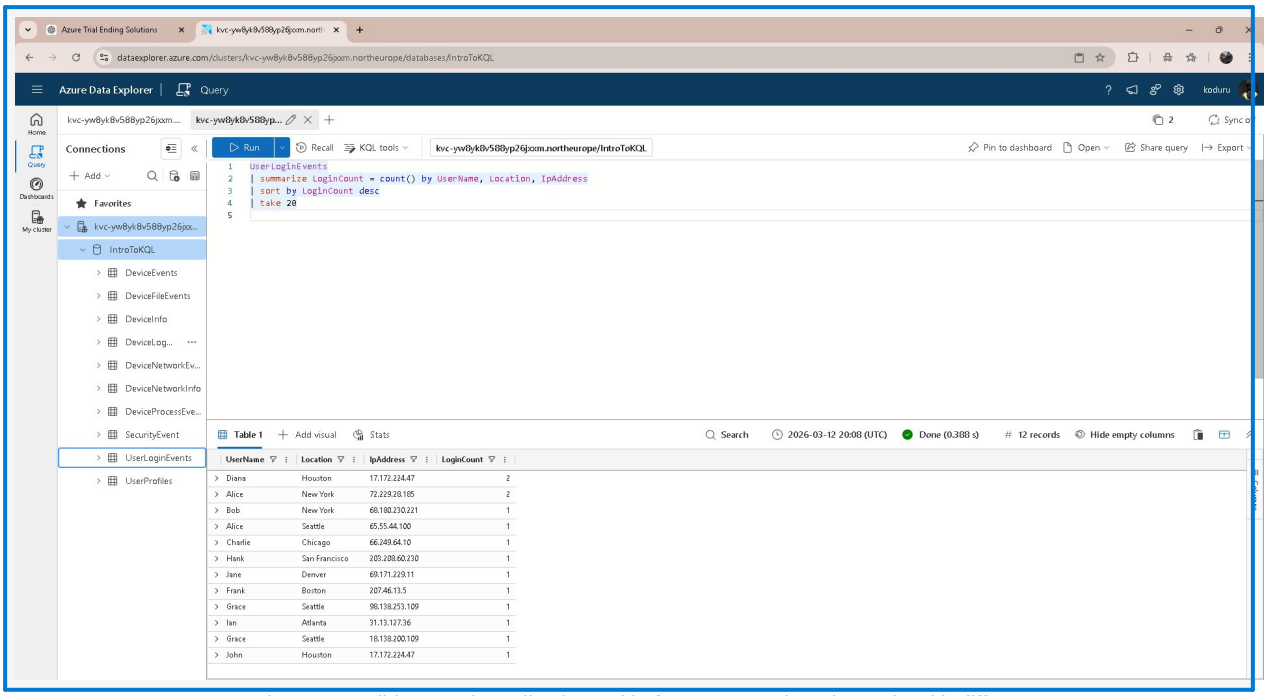
Query 4 — Impossible Travel / Login Anomaly

Detects users logging in from multiple geographic locations. Result: Alice logged in from both New York and Seattle — impossible travel indicator.

```
UserLoginEvents
| summarize LoginCount = count() by UserName, Location, IPAddress
| sort by LoginCount desc
| take 20
```

SOC Use Case: Account takeover, impossible travel alert, credential theft



Screenshot: Impossible Travel — Alice logged in from New York and Seattle with different IPs

6. Resume Project Description

Microsoft Sentinel SIEM Deployment & Threat Detection Lab

- Deployed Microsoft Sentinel SIEM on Azure with a dedicated Log Analytics workspace (sentinal-workspace) and resource group in Central India
- Training Lab Solution provisioned 8 data connectors including MDE, MDI, Cisco Umbrella; manually reconfigured disconnected Azure Activity connector
- Simulated real-world attack scenarios generating 3 live incidents (1 High, 2 Medium severity) via Microsoft Sentinel Training Lab Solution
- Wrote and tested 4 KQL detection queries on Azure Data Explorer covering brute force, LOLBins, C2 beaconing, and impossible travel scenarios
- Explored Threat Intelligence, MITRE ATT&CK; mapping, Analytics rules, Hunting queries, and Incident management workflows